

CE 310 — Week 1 Lab: The Flaw of Averages

Tucson International Airport, NOAA Hourly Weather Data — 2023 (8,760 Hours)

Before You Begin

Download both files from D2L into the same folder on your computer:

- Week1_Tucson_Weather_2023.csv — the raw dataset (8,760 rows)
- Week1_Answer_Sheet.xlsx — your submission template

Answer Sheet layout — six columns, most pre-filled:

Col	Role	You fill this?
A	Question label	No — pre-filled
B	Question description	No — pre-filled
C	Your Excel formula (as text)	Yes
D	Intermediate value (starred questions only)	Yes, if starred
E	Your numerical answer	Yes
F	Unit	No — pre-filled

Example — how a completed row looks:

Col A	Col B	Col C	Col D	Col E	Col F
A1	Annual avg temp	'=AVERAGE(B2:B8761)	—	72.02	°F
A8	Hottest month	'=AVERAGEIF(...)	94.26	7	month #

Type your formula as text. In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, '=AVERAGE(B2:B8761). Without the apostrophe Excel evaluates it and your method is lost. Screenshots are not accepted; the typed formula is what earns the approach credit.

Excel Function Reference

What you want	Function
Mean	=AVERAGE(range)
Median	=MEDIAN(range)
Mode (most common value)	=MODE(range)

What you want	Function
Maximum	=MAX(range)
Minimum	=MIN(range)
Percentile — k=1 for Q1, k=3 for Q3	=QUARTILE(range, k)
Sample standard deviation	=STDEV(range)
Count rows matching one condition	=COUNTIF(range, ">95")
Count rows matching two conditions	=COUNTIFS(col1, ">80", col2, ">50")
Mean for rows matching a condition	=AVERAGEIF(crit_col, value, avg_col)
Nth largest value	=LARGE(range, n)
Month number from a date	=MONTH(date_cell)
Cond. std dev — all Excel (Ctrl+Shift+Enter)	=STDEV(IF(month_col=7, data_col))
Cond. std dev — Excel 365	=STDEV(FILTER(data_col, month_col=7))

Setup — Import the Data (~10 min)

1. Open a new Excel workbook (separate from your Answer Sheet).
2. Go to Data -> Get Data -> From Text/CSV, select Week1_Tucson_Weather_2023.csv, and click Load.
3. Confirm you have 8,760 data rows (plus one header row).
4. Columns: DateTime, Temp_F, Wind_Speed_mph, Relative_Humidity_pct.
5. In column E, add the header Month. In E2, enter =MONTH(A2), then fill down to row 8,761.

SECTION A — Fundamentals (~25 min)

Round temperatures to 2 decimal places. Round counts to whole numbers.

A1 — Annual Average Temperature

Compute the mean of all 8,760 hourly temperature readings.

A2 — Annual Maximum Temperature

Compute the single highest temperature recorded in 2023.

A3 — Annual Minimum Temperature

Compute the single lowest temperature recorded in 2023.

A4 — Annual Temperature Range

Compute MAX – MIN across all 8,760 hours.

A range of ~84°F against an average of ~72°F — the dataset spans from near-freezing to desert-summer extremes.

A5 — Annual Standard Deviation of Temperature

Compute the sample standard deviation of all hourly temperature readings.

A standard deviation of ~18°F against a mean of ~72°F is the first warning that “design for the average” will fail.

Once you have A1 and A5, compute $CV = (A5 \div A1) \times 100\%$ on the side — you’ll cite this in your memo.

A6 — Hours Above 95°F

Count the hours where temperature exceeded 95°F.

A7 — Hours Above 100°F

Count the hours where temperature exceeded 100°F.

(A8 requires a helper table — see end of this section. Skip for now and return after A13.)

A9 — Annual Median Temperature

Compute the median of all 8,760 hourly temperatures.

Compare to A1. Similar values suggest a symmetric distribution; a notable gap suggests skew.

A10 — Q1: 25th Percentile Temperature

Compute the first quartile of all hourly temperatures.

A11 — Q3: 75th Percentile Temperature

Compute the third quartile of all hourly temperatures.

A12 — Interquartile Range (IQR)

Compute $Q3 - Q1$.

A13 — Mode (Most Common Temperature)

Compute the mode of all 8,760 hourly temperatures.

A8 — Month with Highest Average Temperature

Find which calendar month (1–12) had the highest average temperature.

Build a small two-column helper table in a blank area of your workbook — month numbers in one column, monthly averages using AVERAGEIF in the other. Then identify which row has the maximum.

Figure 1 — Helper table structure:

Month	Avg Temp (°F)
1	=AVERAGEIF(\$E\$2:\$E\$8761, 1, \$B\$2:\$B\$8761)
2	=AVERAGEIF(\$E\$2:\$E\$8761, 2, \$B\$2:\$B\$8761)
...	...
12	=AVERAGEIF(\$E\$2:\$E\$8761, 12, \$B\$2:\$B\$8761)

Lock the ranges with \$ so they don't shift as you fill down.

- Col D (starred): Average temperature of the hottest month
- Col E: Month number (1–12)

SECTION B — In-Depth Analysis

B1 — ASHRAE 1% Design Dry-Bulb Temperature

The cooling design temperature is defined as the 88th-highest hourly temperature of the year (1% of 8,760 hours = 87.6, rounded to 88).

Use the Function Reference to find the appropriate function.

B2 — Design Temperature Minus Annual Average

Subtract A1 from B1. This is how many degrees an average-based HVAC system would be undersized.

B3 (starred) — Standard Deviation of Temperature in July

How much does temperature vary hour-to-hour within July? STDEV alone cannot filter by month — use a conditional std dev from the Function Reference.

Two options — try Excel 365 first; if it returns #NAME?, use the all-Excel version:

- Excel 365: =STDEV(FILTER(B2:B8761, E2:E8761=7)) — press Enter normally
- All Excel: =STDEV(IF(E2:E8761=7, B2:B8761)) — press Ctrl+Shift+Enter (not just Enter)

When the array formula works correctly, curly braces appear automatically around it:

Before: =STDEV(IF(E2:E8761=7, B2:B8761))

After: {=STDEV(IF(E2:E8761=7, B2:B8761))}

If you typed the curly braces yourself, the formula will not work — they must appear on their own after Ctrl+Shift+Enter.

- Col D (starred): July average temperature — confirms you filtered the right month
- Col E: July standard deviation (°F)

B4 (starred) — Standard Deviation of Temperature in January

Same approach as B3, for January (month = 1).

- Col D (starred): January average temperature
- Col E: January standard deviation (°F)

Which month is more variable? Think about what drives that variability.

B5 — Average Temperature in July

Compute the average temperature for July hours only.

B6 — Average Temperature in January

Compute the average temperature for January hours only.

Questions B7–B11 — Wind Loads (ASCE 7-22)

The same flaw appears in structural engineering. The formula $P = 0.00256 \times C_a \times V^2$ governs wind pressure on every building and sign structure. Designing for average wind speed instead of peak produces dangerously undersized structures.

B7 — Annual Average Wind Speed

Compute the mean of all 8,760 wind speed readings (Column C).

B8 — Annual Maximum Wind Speed

Compute the single highest wind speed recorded in 2023.

B9 (starred) — Wind Pressure at Annual Average Speed

$P = 0.00256 \times C_a \times V^2$ where $C_a = 1.3$, V in mph, P in psf.

Apply this to your average speed from B7.

- Col D (starred): V^2 (your B7 value, squared)
- Col E: P in psf, rounded to 3 decimal places

B10 (starred) — Wind Pressure at Annual Maximum Speed

Same formula, using your maximum speed from B8.

- Col D (starred): V^2 (your B8 value, squared)
- Col E: P in psf, rounded to 3 decimal places

B11 (starred) — Ratio of Peak to Average Wind Pressure

Divide B10 by B9.

- Col D (starred): Your B10 value (peak pressure)
- Col E: The ratio, rounded to 2 decimal places

This ratio is how many times the design load a billboard sized for average wind would experience during the peak gust.

Questions B12–B14 — Latent Load (ASHRAE 62.1)

HVAC systems must remove moisture from outdoor air. An engineer who sizes dehumidification for the annual average humidity instead of peak monsoon conditions will produce a system that fails when it is needed most.

B12 — Annual Average Relative Humidity

Compute the mean of all 8,760 relative humidity readings (Column D).

B13 — Hours Where Relative Humidity Exceeds 60%

Count the hours where outdoor RH exceeded 60%.

Tucson's North American Monsoon pushes RH above 60% for over 1,000 hours per year — a desert city with a real dehumidification load.

B14 — Hours with Warm Temperatures AND High Humidity

Count the hours where temperature exceeded 80°F and relative humidity exceeded 50% simultaneously. Refer to the Function Reference for the two-condition count function.

Target shape for Chart 1

Figure 1: Target shape for Chart 1

Target shape for Chart 2

Figure 2: Target shape for Chart 2

Required Charts

Add both charts to your working Excel file (not the Answer Sheet).

Chart 1 — Monthly Average Temperature

Use your A8 helper table data. Plot a line or column chart: months 1–12 on the X-axis, monthly average temperature on the Y-axis. Add a horizontal reference line at the annual average (A1). Title: Monthly Average Temperature — Tucson, AZ 2023.

Chart 2 — Hourly Temperature vs. Annual Average: July 1–7

Step 1 — Extract 168 rows for July 1–7:

1. Click any cell in your data table -> Data -> Filter.
2. Click the dropdown on DateTime -> Date Filters -> Between.
3. Enter 7/1/2023 and 7/7/2023 -> OK. You should see 168 rows.
4. Copy the filtered Temp_F column -> paste (Values only) into a new blank sheet.
5. Turn the filter off: Data -> Filter again.

(If date filters don't work in your Excel version, ask your preceptor for the helper-column method.)

Step 2 — Build the chart:

1. Add a column Annual_Avg next to the pasted temperatures, filled with your A1 answer.
 2. Select both columns -> Insert -> Line Chart.
 3. Format Annual_Avg as a dashed red line.
 4. Title: Hourly Temperature vs. Annual Average — Tucson, July 1–7, 2023. Label both axes.
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Memo Deliverable

Type your memo in the Written sheet of the Answer Sheet (yellow cells P1, P2, P3).

To: Building Systems Design Client From: [Your Name] Subject: Why Annual Average Temperature Is Insufficient for HVAC Design — Tucson, AZ

Paragraph 1 — The averages problem. State the annual average (A1), annual maximum (A2), and CV ($= A5 \div A1 \times 100\%$). Use the CV value and the three-tier threshold from lecture ($< 10\%$ reliable / $10\text{--}20\%$ report σ / $> 20\%$ use percentiles) to explain whether the mean alone is a reliable design value for Tucson's temperature. Reference B2 for the gap.

Paragraph 2 — The design temperature. State the ASHRAE design temperature (B1) and hours above 100°F (A7). Include Chart 2.

Paragraph 3 — Humidity burden. Using B12 and B13, explain why annual-average humidity understates the dehumidification load in Tucson.

Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet
 - ☐ File saved as .xlsx and named CE310_W1_<NetID>.xlsx
 - ☐ Sheet tabs not renamed, reordered, or deleted
 - ☐ Every answer typed into its designated cell in Col E — answers placed anywhere else receive no credit
 - ☐ Your formula typed as text in Col C, with an apostrophe before the = — screenshots are not accepted
 - ☐ ★ questions also have their intermediate value in Col D
 - ☐ Uploaded to D2L before the deadline
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Grading Summary

Component	Points
Section A: A1–A13	16
Section B: B1–B14	60
Chart 1 — Monthly averages	7
Chart 2 — July hourly vs. annual average	8
Memo — 3 paragraphs referencing specific values	9
Total	100